

CAPITAL APPROVAL REQUEST

Project TSF2 PH 1&2 Dam Wall Raise
Operation Perkoa Mine
Project Owner [REDACTED]
Project Sponsor [REDACTED]
Date: 25th May 2020
Revision: 1.0

Request for approval:

For funding to raise the existing TSF2 Phase 1&2 by 0.75m to increase the capacity of TSF2 Ph 1&2 to allow the mine to operate at full production until 7th August 2020.

This will allow sufficient time to complete the Construction of TSF2 Phase3

The cost for the Dam Wall raise is estimated at US\$381,698.50

Situation:

The Capacity of the TSF2 PH1&2 is being depleted at a rate of 48 000T/m = 28742.5m³/m

The remaining capacity left as of the 2nd April 2020 is:-

Current Situation		11-Feb-20	24-Mar-20	Change
Total Initial Volume	m3	998100	998100	
1 m freeboard Volume	m3	908007	908007	
Water Volume	m3	383173	245735	137438
Solids Volume	m3	352062	411190	-59128
Remaining Volume	m3	172772	251082	-78310

At the current mining and processing rate of 48 000T/m the TSF2 PH1&2 will run out of storage capacity by 7th August 2020.

With the upcoming rainy season, is highly unlikely that the completion of TSF2 PH3 will be completed before TSF2 PH1&2 runs out of capacity.

The one option is to construct TSF2 PH3 with 3 Cells, adding to the Cost of construction and not adding any value.

Original Opportunity identified 0.75m wall raise:

Original Proposed lift

To raise TSF2 Ph1&2 by 0.75m

This will give an additional capacity of

$$215\,504.5\text{m}^2 \times 0.75\text{m} = 161\,628.4\text{m}^3 \times 1.67\text{sg} = 269\,918\text{T} / 48\,000\text{T/m} = 5.62\text{ months}$$

Estimated Cost was \$193 077 = \$0.715/T of tailings

Revised Proposed Lift 1.2m wall raise

To raise TSF2 Ph1&2 by 1.2m

$$215\,504.5\text{m}^2 \times 1.20\text{m} = 258\,604.8\text{m}^3 \times 1.6767\text{sg} = 431\,870\text{T} / 48\,000\text{T/m} = 8.99\text{ months}$$

This will extend the life for deposition from 7th August 2020 to 7th April 2021

Capital Approval Request

Capital approval request demonstrating the engineering justification for revising the interim berm raise from the initial 0.75 m proposal to the final 1.2 m configuration, increasing operational storage life from approximately 5.6 to 8.99 months and enabling continued mine production during construction of TSF II Phase III.